



United International University
Department of Computer Science and Engineering

CSE 3313: Computer Architecture

Assignment

Total Marks: 5

Q1.

You have to modify the block diagram for the single-cycle data path and write the control unit values for this instruction: 'powerload \$s0, \$s1, \$s2'. The job of the given instruction is to read the value of the source registers (rs, rt) and calculate twice the value of each of them. Assuming variables a and b are in register rs, and rt, the task is to calculate the values of a multiplied by 4 and b multiplied by 2. After that, the calculated values will be added first to get an address. We need to read the value of that address and write it to the destination register rd. Afterwards, copy the value at source register rt (which is \$s2 in this instruction and place it on Program Counter (PC).) The machine code format for the given instruction is given below in table 1, where rs, rt, and rd will contain the index numbers of \$s1, \$s2, and \$s0 registers, respectively.

Opcode	Rs	Rt	Rd	Shamt	Function Code
6 bits	5 bits	5 bits	5 bits	5 bits	6 bits

Table 1: Machine code format of instruction powerload